

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 2.5 RATES OF REACTIONS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the Disappearing Cross. Mwalimu Achieng's class in Nakuru County has been asking: Why do some chemical reactions happen in a flash while others take days, and how can we control the speed of reactions to make life in Kenya safer, healthier and more productive? Use your science notebook, the class data tables and your Driving Question Board (DQB) to help you. Write your answers in the space under each prompt. Use full sentences, include numbers with units where you can, and use the words particles, effective collisions and activation energy. Remember the claim, evidence and reasoning structure. In Section 3 you will use a table of results.

Section 1: Explain the Disappearing Cross (Claim, Evidence, Reasoning)

In the anchor demonstration, 10 cm³ of 1.0 mol/dm³ hydrochloric acid was put in each of two beakers and 50 cm³ of 0.15 mol/dm³ sodium thiosulfate was added. Beaker A was at 10 C and Beaker B was at 50 C. The black cross disappeared at 35 s in one beaker and at 270 s in the other.

(a) Write a CLAIM that answers Wanjiru's question: why did one beaker reach the end point faster than the other?

(b) Give EVIDENCE from the demo: the times, how many times faster, and the rate for each beaker using rate = 1/t.

(c) Write your REASONING using particles, collisions, kinetic energy and activation energy.

(d) State what was the same in both beakers, and what you can say about the amount of sulfur

(a) Claim: Warm Beaker B (50 C) reacted faster than cold Beaker A (10 C) because the higher temperature made more of the collisions between particles effective.

(b) Evidence: The cross vanished at 35 s in Beaker B and at 270 s in Beaker A. The difference is $270 - 35 = 235$ s, and $270/35 = 7.7$, so Beaker B was 7.7 times faster. Rate of B = $1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$. Rate of A = $1/270 = 3.7 \times 10^{-3} \text{ s}^{-1}$.

(c) Reasoning: At 50 C the particles have more kinetic energy. They move faster, so they collide more often and with more energy. A larger fraction of the collisions have at least the activation energy (the minimum energy needed for a collision to react), so there are more effective collisions per second. More effective collisions per second means sulfur forms faster and hides the cross sooner. At 10 C the particles have less energy, so fewer collisions are effective and the cross stays visible longer.

(d) The chemicals, amounts (50 cm³ of thiosulfate, 10 cm³ of HCl), beaker type, cross and observer were the same; only temperature differed. Both beakers end with the same amount of sulfur, because temperature changes the speed of the reaction, not the final amount made.

made at the end.

Section 2: Fair Testing and Measuring Rate

Amina wants to test whether temperature changes the rate, using the cross method.

(a) Name the independent variable, the dependent variable and at least three control variables for her test.

(b) Amina says: 'I will use a different observer and a bigger beaker for the hot test, because it is faster anyway.' Explain why this plan is not a fair test.

(c) Explain why a shorter time for the cross to disappear gives a bigger rate.

(d) Name two other ways, besides the time for a precipitate to hide a cross, that the rate of a reaction can be measured.

(a) Independent variable: temperature (the factor you change). Dependent variable: the time for the cross to disappear (then the rate, $1/t$). Control variables (any three): 50 cm³ of 0.15 mol/dm³ sodium thiosulfate, 10 cm³ of 1.0 mol/dm³ HCl, the same type of beaker, the same cross, the same observer, the same lighting.

(b) It is not fair because more than one thing would change: temperature, observer and beaker size. A different observer may judge the cross as gone at a different time, and a different beaker changes the depth of liquid. We could not tell whether temperature or the other changes caused the different time. Only the independent variable should change.

(c) Rate = $1/t$. When t is small, 1 divided by t is a bigger number. For example, $1/35 = 28.6 \times 10^{-3} \text{ s}^{-1}$ is bigger than $1/270 = 3.7 \times 10^{-3} \text{ s}^{-1}$, so the faster reaction has the bigger rate.

(d) Measure the volume of gas produced over time (rate = volume in cm³ divided by time in s), and measure the loss of mass as a gas escapes.

Section 3: Using Evidence from the Tables

Here is the class-mean data from Lessons 3 and 4. All used 10 cm³ of 1.0 mol/dm³ HCl and a total thiosulfate volume of 50 cm³.

Temperature test (0.15 mol/dm³ thiosulfate):

Temperature (C)	Time for cross to disappear (s)	Rate ($\times 10^{-3} \text{ s}^{-1}$)

(a) The rate increases as temperature rises: 3.7, 9.1, 19.2 and $38.5 \times 10^{-3} \text{ s}^{-1}$ at 10, 25, 40 and 55 C. From 10 C to 55 C the rate rises from 3.7 to 38.5, about ten times bigger.

(b) The demo time of 35 s at 50 C lies between 52 s at 40 C and 26 s at 55 C, so it fits the pattern: the higher the temperature, the shorter the time.

(c) As concentration falls from 0.15 to 0.06 mol/dm³ the time rises from 110 s to 275 s and the rate falls from 9.1 to $3.6 \times 10^{-3} \text{ s}^{-1}$. Rate divided by concentration is about 60 in every row (for example $9.1/0.15$ is about 61 and $3.6/0.06 = 60$), so rate is roughly proportional to concentration.

(d) A higher concentration has more particles in the same volume, so particles collide more often and there are more effective collisions per second. The temperature was 25 C in every test, so the particles had the same kinetic energy and the same activation energy requirement; the energy of the particles did not change.

10	270	3.7
25	110	9.1
40	52	19.2
55	26	38.5

Concentration test (all at 25 C):

Sodium thiosulfate (mol/dm³)	Time (s)	Rate (x10⁻³ s⁻¹)
0.15	110	9.1
0.12	138	7.2
0.09	183	5.5
0.06	275	3.6

(a) Describe the pattern in rate as temperature rises from 10 C to 55 C.

(b) The anchor demo at 50 C took 35 s. Show whether this fits your pattern in (a).

(c) Describe the pattern between concentration and rate, and use the numbers to say how rate and concentration are related.

(d) Explain the concentration pattern using particles. Why does the energy of the particles not explain it?

Section 4: Surface Area, Catalysts and Light

(a) In Lesson 5, 0.5 g of marble chips (CaCO_3) in 25 cm³ of 1.0 mol/dm³ HCl at 25 °C gave initial rates over the first 30 s of 0.40 cm³/s for large chips, 0.93 cm³/s for small chips and 2.07 cm³/s for powder. Explain why powder gave the fastest rate, and say what happened to the final volume of CO₂ for all three.

(b) In Lesson 6, 10 cm³ of 3% hydrogen peroxide with detergent gave 0 cm of foam with no additive, 0 cm with sand, 0.5 cm with a boiled liver cube and 9 cm with a fresh liver cube after 30 s. Explain why fresh liver made so much foam, using the words catalyst, enzyme and activation energy. Why did the boiled liver not work well?

(c) Kiprono says light is a catalyst. Do you agree? Give a reason and an example.

(a) Powder has a much larger surface area than large chips, so more CaCO_3 particles are exposed to HCl particles. There are more effective collisions per second, so CO₂ is made faster and the volume-time curve is steeper. The mass of marble, the acid and the temperature were the same, so all three reach the same maximum of about 120 cm³ of CO₂; surface area changes the speed, not the final amount.

(b) Fresh liver contains the enzyme catalase, a biological catalyst. It gives the hydrogen peroxide reaction ($2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$) an alternative pathway with a lower activation energy, so a larger fraction of collisions are effective and oxygen forms quickly, making 9 cm of foam. The liver is not used up, so it can work again. Boiling damages the enzyme, so the boiled liver gave only 0.5 cm. Sand gave 0 cm, showing that not every solid speeds up the reaction.

(c) I disagree. Light is a source of energy for some reactions, such as photosynthesis in plant leaves; it supplies energy, whereas a catalyst provides an alternative pathway with lower activation energy and is not used up.

Section 5: Solve a Kenyan Problem and Reflect on Your Model

A trader in Kisumu finds that her fresh fish from Lake Victoria spoils within hours. A farmer in Kericho wants to speed up composting of tea waste.

(a) For the fish trader, choose one way to slow down spoilage reactions. Explain, using particles and collisions, why it works.

(b) For the Kericho farmer, choose one factor to change to speed up composting. Say how

(a) Cool the fish, for example by keeping it in a fridge at 4 °C or on ice, or dry and salt it. Cooling gives the particles less kinetic energy, so they collide less often and with less energy. Fewer collisions reach the activation energy, so fewer collisions are effective per second and the spoilage reactions are slower. (Drying and salting remove the water needed for spoilage reactions.) Fish keeps longer.

(b) Increase the surface area by chopping the tea waste into small pieces, or raise the temperature by keeping the compost warm. Smaller pieces expose more particles, so there are more effective collisions per second and the rate rises. Warmth gives particles more kinetic energy, so more collisions are effective.

(c) Enzymes work best at an optimum temperature and pH where their shape suits the reaction. Amylase works best near 37 °C and pH 7. Far from these conditions the enzyme's shape is damaged, as with boiled liver, so it works slowly or not at all.

(d) At the start I only had a first sketch of particles and could not say why warm Beaker B was faster. Now my model says rate depends on the number of effective collisions per second, and temperature, concentration, surface area and catalysts each change this. Answered DQB question: Why was Beaker B faster than Beaker A? Because at 50 °C particles have more kinetic energy, collide more often and with more energy, so more collisions have at least the activation energy.

(e) Advice: keep food cool, dry or salted to make it last longer, because slowing reactions slows spoilage. A career: a KEMRI

you would change it and explain why it works. (c) Use the idea of an optimum to explain why the enzyme amylase in your body works best near 37 C and pH 7, and what you would expect if conditions were far from this. (d) Describe how your particle model has changed since Lesson 1, and state one DQB question that is now answered. (e) Write one sentence of advice for your community about controlling reaction rates, and name a career that uses this knowledge.	biochemist.
--	-------------

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning for the anchor phenomenon	Gives a clear claim, uses exact data (35 s, 270 s, 7.7 times, rates 28.6 and $3.7 \times 10^{-3} \text{ s}^{-1}$) and links more kinetic energy, more frequent and energetic collisions, and activation energy to more effective collisions per second.	Gives a claim and uses most of the data, with reasoning that mentions collisions and energy but may leave out activation energy or effective collisions.	Claim is vague or reasoning says only that particles move faster or that the reaction is hotter; little or no data is used.
Fair testing and use of rate = $1/t$	Names independent, dependent and control variables correctly, explains why changing several things is unfair, and correctly explains rate = $1/t$.	Names most variables and gives a basic reason for fairness; rate = $1/t$ is used correctly with small gaps in the explanation.	Confuses variables or does not show why the plan is unfair; rate and time are mixed up.
Interpreting data tables	Describes trends accurately with numbers, checks that 35 s fits between 52 s and 26 s, and shows rate roughly proportional to concentration (about 60).	Describes trends correctly but with few numbers or without testing proportionality or the 35 s link.	Misreads the tables or states trends without evidence.
Understanding of surface area, catalysts and light	Explains surface area, catalysts (lower activation energy, not used up, enzyme) and light correctly, and states that factors change speed, not the final amount.	Explains most factors correctly with small gaps, such as missing activation energy or the final amount.	Mixes up catalysts and light or gives no particle explanation.
Applying the model to Kenyan contexts and reflecting	Chooses suitable factors for the fish and compost problems, explains them with	Suggests suitable actions with partial particle explanations and a general	Gives actions without reasons, or no reflection on how the model changed.

	collision theory, links to optimum conditions for enzymes, and clearly compares the model with Lesson 1.	reflection.	
--	--	-------------	--

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.
